

Django Backend Setup — Step by Step

Python & VS Code environment · Project structure · Django + DRF + MongoDB

1 Install on Your System

- **Python 3.11 or 3.12** → python.org → during install, check "**Add Python to PATH**" (Windows)
- Verify: `python --version` and `pip --version`
- **Git** → git-scm.com (if not already installed)
- **VS Code** → code.visualstudio.com
- **MongoDB Atlas** account (free tier, cloud) → create a cluster → copy the connection string. No local Mongo install needed.
- (Optional) **MongoDB Compass** — GUI to browse your Atlas data

2 VS Code Extensions

- **Python** (Microsoft)
- **Pylance**
- **Django** (Baptiste Darthenay) — template/keyword highlighting
- **Thunder Client** — test your APIs inside VS Code
- **MongoDB for VS Code** — browse Atlas collections from the editor
- **DotENV** — `.env` syntax highlighting

3 Project Folder + Virtual Environment

```
mkdir backend && cd backend
python -m venv venv

# activate:
venv\Scripts\activate      # Windows
source venv/bin/activate   # Mac/Linux

pip install --upgrade pip
```

4 Install Django & Packages

```
pip install django djangorestframework djangorestframework-simplejwt django-cors-headers python-decouple pymongo
pip freeze > requirements.txt
```

5 Start the Project

```
django-admin startproject config .
```

6 Create the Apps (matching your architecture)

```
mkdir apps
type nul > apps/__init__.py      # Windows
# touch apps/__init__.py        # Mac/Linux

python manage.py startapp authentication apps/authentication
python manage.py startapp tickets apps/tickets
python manage.py startapp agents apps/agents
python manage.py startapp admin_panel apps/admin_panel
python manage.py startapp history apps/history
python manage.py startapp reports apps/reports

mkdir ai_engine
```

Final folder structure:

```
backend/
├── config/          # settings.py, urls.py
├── apps/
│   ├── authentication/
│   ├── tickets/
│   ├── agents/
│   ├── admin_panel/
│   ├── history/
│   └── reports/
├── ai_engine/       # classification/priority/similarity logic (added later)
├── .env
├── requirements.txt
└── manage.py
```

7 .env File (project root)

```
SECRET_KEY=your-secret-key-here
DEBUG=True
MONGO_URI=mongodb+srv://<user>:<password>@<cluster>.mongodb.net/ticketdb
```

8 config/settings.py — Key Edits

```
from decouple import config

SECRET_KEY = config('SECRET_KEY')
DEBUG = config('DEBUG', cast=bool)

INSTALLED_APPS = [
    ...
    'rest_framework',
    'rest_framework_simplejwt',
    'corsheaders',
    'apps.authentication',
    'apps.tickets',
    'apps.agents',
    'apps.admin_panel',
    'apps.history',
    'apps.reports',
]

MIDDLEWARE = [
    'corsheaders.middleware.CorsMiddleware',  # keep near the top
    ...
]

CORS_ALLOWED_ORIGINS = [
    "http://localhost:5173",  # your Vite React dev server
]

REST_FRAMEWORK = {
    'DEFAULT_AUTHENTICATION_CLASSES': (
        'rest_framework_simplejwt.authentication.JWTAuthentication',
    ),
}

from datetime import timedelta
SIMPLE_JWT = {
    'ACCESS_TOKEN_LIFETIME': timedelta(hours=1),
    'REFRESH_TOKEN_LIFETIME': timedelta(days=7),
}
```

9 MongoDB Connection (config/db.py)

Since Django's ORM doesn't talk to MongoDB natively, connect via PyMongo directly and use DRF serializers (not `ModelSerializer`) for tickets/users:

```
from pymongo import MongoClient
from decouple import config

client = MongoClient(config('MONGO_URI'))
db = client.get_default_database()

tickets_collection = db['tickets']
users_collection = db['users']
```

Import `db` wherever a view needs to read/write — e.g. `apps/tickets/views.py` uses `tickets_collection.insert_one(...)`.

10 Run It

```
python manage.py runserver
```

Visit `http://127.0.0.1:8000` — you should see Django's default page. Test an endpoint with Thunder Client once you've written your first view.

11 Connect to Your React Frontend

In your React app's `.env` or Axios config, point to:

```
VITE_API_URL=http://127.0.0.1:8000/api
```

Next step: build the `authentication` app first (register/login/JWT), since every other app depends on it.

— End of Django Backend Setup Guide —